exercise 5.

problem 1 (Graph and Tree Definitions) Prove that the following definitions are all equivalent.

To prove the seven equivalences, we can first prove (1) → (3) → (5) → (1) and (1) → (4) → (6) → (1), (1) ↔ (2) (1) ↔ (7).

Method of use //

A connected graph on V vertices has at least V-1 edges. An acyclic graph on V vertices has at most V-1 edges.

proof:

By induction on V. Base case V=1: 0 edges = V-1.

For the inductive step: a connected graph has a spanning three with exactly V-1 edges (remove cycle edges repeatedly). An acyclic graph (forest) with k components has exactly V-k edges, so at most V-1.

I: (1) → (3) → (5) → (1).

(1) → (3) Connected + acyclic ⇒ connected + $E \le V-1$

If G is connected and acyclic, it is a forest with 1 component, so $E = V-1 \le V-1$ ※

(3) → (5): Connected + $E \le V-1$ ⇒ acyclic + $E \ge V-1$

A connected graph has $E \ge V-1$ (Method of use). Combined with $E \le V-1$, we get $E = V-1$. A connected graph with exactly V-1 edges must be acyclic (if a cycle existed, removing one cycle edge keeps connectivity with V-2 edges, contradicting the method of use). So acyclic with $E = V-1 \ge V-1$ ※

(5) → (1):

Acyclic + $E \ge V-1$ ⇒ connected + acyclic

An acyclic graph with k components has $E = V-k$. Since $E \ge V-1$, we get $V-k \ge V-1$, so $k \le 1$. Since $k \ge 1$, we have $k = 1$ → connected. ※

Therefore. I: (1) → (3) → (5) → (1) is true.

II : (1) → (4) → (6) → (1)

$\boxed{(1) \to (4)}$: connected + acyclic ⇒ minimally connected

Take any edge uv, Since G is acyclic, the only path from u to v is the edge uv itself. Removing uv disconnects u from v. Hence G is minimally connected ※

$\boxed{(4) \to (6)}$: Minimally connected ⇒ maximally acyclic

Minimally connected ⇒ connected and acyclic (if a cycle existed, removing a cycle edge keeps connectivity, contradicting minimality). For any non-edge uv, Since G is connected, a path from u to v exists; adding edge uv creates a cycle. Hence maximally acyclic ※.

$\boxed{(6) \to (1)}$: Maximally acyclic ⇒ connected + acyclic

Acyclic by definition. Suppose not connected:

$\exists\, u, v$ in different components. Adding edge uv

cannot create a cycle (no path between them). Contradicting "maximally acyclic." Hence connected ※.

Therefore II : (1) → (4) → (6) → (1) is true.

Equivalence : (1) ⟷ (2)

(1) → (2) :

A connected acyclic graph is itself a forest (acyclic), and being connected, it is exactly one component of that forest //

(2) → (1) :

One component of a forest is a maximal connected subgraph of an acyclic graph. A subgraph of an acyclic graph is acyclic; a component is connected. Hence connected acyclic graph //

Therefore Equivalence : (1) ⟷ (2) is true.

Equivalence : (1) $\Leftrightarrow$ (7)

(4) $\rightarrow$ (7) :

Connected $\Rightarrow$ path exists between every pair. If two distinct paths existed between u and v, their union contains a cycle (traverse one forward, the other backward), contradicting acyclicity. So the path is unique //

(7) $\rightarrow$ (4) :

Unique path $\Rightarrow$ connectivity (a path exists). Also acyclic : if a cycle $V_1 \rightarrow V_2 \rightarrow \dots \rightarrow V_k \rightarrow V_1$ existed, then V_1 to V_2 would have two distinct paths (direct edge + going around), contradicting uniqueness ※

Therefore Equivalence (4) $\Leftrightarrow$ (7) is true.

To Conclusion All seven definitions are equivalent.

Problem 2. Sparse representation of graphs.

Graph 1.

CSC Data:

col_pointers : [0, 2, 5, 8, 11, 12]

row_indices : [1, 2, 0, 2, 3, 0, 1, 3, 1, 2, 4, 3]

values : [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]

CSC Decoding

Column	Range	row-indices	Edges into this column
0 (A)	[0, 2)	[1, 2]	B-A, C-A
1 (B)	[2, 5)	[0, 2, 3]	A-B, C-B, D-B
2 (C)	[5, 8)	[0, 1, 3]	A-C, B-C, D-C
3 (D)	[8, 11)	[1, 2, 4]	B-D, C-D, E-D
4 (E)	[11, 12)	[3]	D-E

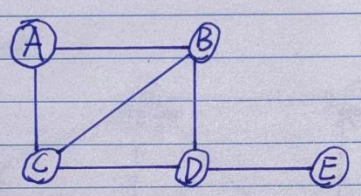
(a) Adjacency Matrix

A	0	1	1	0	0
B	1	0	1	1	0
C	1	1	0	1	0
D	0	1	1	0	1
E	0	0	0	1	0

The matrix is symmetric since the graph is undirected.

(b) Graph Diagram

Undirected.



Edge list {A-B, A-C, B-C, B-D, C-D, D-E}.

Total 6 edges, 5 vertices.

Graph 2.

CSC Data

Col-pointers : [0, 0, 2, 4, 5, 7]

row-indices : [0, 3, 0, 1, 2, 1, 3]

values : [1, 1, 1, 1, 1, 1, 1]

CSC Decoding

column	Range	row-indices	Edges into this column
0 (A)	[0, 0)	[ ]	No edges into A
1 (B)	[0, 2)	[0, 3]	A→B, D→B
2 (C)	[2, 4)	[0, 1]	A→C, B→C
3 (D)	[4, 5)	[2]	C→D
4 (E)	[5, 7)	[1, 3]	B→E, D→E

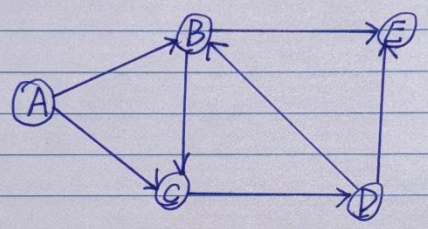
a) Adjacency Matrix

A	0	1	1	0	0
B	0	0	1	0	1
C	0	0	0	1	0
D	0	1	0	0	1
E	0	0	0	0	0

The matrix is asymmetric since the graph is directed.

b) Graph Diagram

Directed.



Directed edge list : A→B , A→C, B→C , B→E , C→D, D→B, D→E.

Total : 7 directed edges.

c) Unique Cycle in the Directed Graph.

B→ C→ D→B

Verification

Edge	Matrix entry	Exists ?
B→C	M[1][2] = 1	exist
C→D	M[2][3]=1	exist
D→B	M[3][1]=1	exist.

Vertex A has in-degree 0 (no incoming edges), so A cannot appear in any cycle. vertex E has out-degree 0 (no outgoing edges), so E cannot appear in any cycle. The only remaining vertices are {B, C, D}, and the cycle B→ C→ D→ B is the unique directed cycle.